

A profile-integral maximum-degree bound for diameter-2 edge-critical graphs

Reviewer edition 1 - candidate general structural theorem

Paul Lenz research project; mathematical development and drafting with ChatGPT/Geeps

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Abstract

We present the current candidate profile-integral argument proving the profile-integral surplus bound in the selected/residual framework and deriving an exact rational maximum-degree threshold which, for orders at least six, forces a strict improvement over the balanced complete-bipartite edge bound. The argument is a general structural candidate rather than a proof of the unrestricted Murty-Simon conjecture. Its finite integer checks are retained and replayed internally; independent mathematical review and novelty assessment remain open.

Reviewer status. complete candidate hand argument; independent review and novelty assessment OPEN. This reviewer edition is an editorial rendering of the canonical source proof listed below. It does not convert same-assistant checking into external independence, does not make a novelty or priority claim, and does not claim the unrestricted Murty-Simon conjecture unless the source proof itself proves such a statement.

Canonical claim. $n \geq 6$ and $\Delta(G) \geq (293/500)n$ imply $e(G) < \text{floor}(n^2/4)$.

Canonical proof source. `project/research/general_n/2026-09-08-profile-integral-v1/PROOF.md`. The source is reproduced in full below; substantive mathematical changes must be made in the canonical proof first and then rebuilt into this edition.

A profile-integral bound and the 293/500 maximum-degree candidate

8 September 2026. Paul Lenz: research direction. ChatGPT/Geeps: derivation, software, drafting and internal audit.

Status: complete candidate hand argument with explicit finite integer certificates. Internal computations REPRODUCED; independent mathematical review and novelty assessment OPEN. Not PROJECT-CERTIFIED, not a full Lean proof, and not a solution of the unrestricted Murty-Simon conjecture.

Statement and scope

Let G be a finite simple diameter-two edge-critical graph, with n vertices, m edges and maximum degree b . Put $a=n-1-b$ and $t=m-b(n-b)$. The argument below gives, for $a \geq 1$,

$$t < a^2/24 + a/8. \tag{P}$$

Together with explicitly bounded small cases, it gives the candidate implication

$$n \geq 6 \text{ and } b \geq (293/500)n \implies m < \text{floor}(n^2/4). \tag{T}$$

Here $293/500=0.586$ exactly. The strict statement cannot be extended to all $n \geq 4$: $K(2,3)$ has $n=5$, $b=3$ and $m=6=\text{floor}(25/4)$, while $3 \geq 293 \cdot 5/500$.

This completes the previously unfinished 0.586 route. The original Jensen step is valid, but unnecessary: a pointwise scalar bound can be summed directly. The proof also avoids dependence on the older cubic residual inequality or its $4/81$ coefficient. It does depend on the graph-to-demand construction and exact threshold count, which are rederived below. No positive-surplus graph is assumed to exist or found by this work.

1. Graph-to-demand construction

Write H for the complement of G . An adjacent total-dominating pair in H is a pair nonadjacent in G with no common G -neighbour. Thus H has none; inserting any missing H -edge creates one, by edge-criticality of G .

Choose v of minimum H -degree a and put $A=N_H(v)$, $B=V(H)$ minus $N_H[v]$, so $|A|=a$ and $|B|=b$. Insert a missing pair uw inside $H[B]$. A newly created adjacent total-dominating pair must use u or w : otherwise neither its adjacency nor its neighbourhood union changed. The pair $\{u,w\}$ itself misses v . Consequently a new pair uses exactly one endpoint, say u , and another vertex i . Its edge ui already existed. To dominate v it must have i in A . Before insertion its open H -neighbourhood union was exactly $V(H)$ minus $\{w\}$; otherwise it would already have totally dominated H . In particular, both u and i miss w . Denote this quasi-edge by $ui \rightarrow w$.

Choose one such cross-edge for each missing unordered B -pair. Two distinct missing pairs cannot yield the same selected cross-edge: its B -endpoint and its unique undominated vertex recover the pair. At a fixed source, selected labels and supplements are separately distinct. Every unordered B -pair carries at most one selected orientation. All remaining $H[A,B]$ edges are called residual.

Let ρ_u and R_i be residual degrees at u in B and i in A , respectively, with common total r . Put $F=\text{complement}(H[A])$, $d_i=\deg_F(i)$, and let x_i be the number of actual selections at label i . Define

$$s_i = \max(0, d_i - R_i), \quad S = \sum_i s_i.$$

The selected cross-edges together with $H[B]$ account for $\text{binom}(b,2)$ edges. Counting H gives

$$e(F) = r + t, \quad \sum_i d_i = 2(r + t), \quad \sum_i R_i = r.$$

Minimum H -degree gives $a - d_i + R_i + x_i \geq a$. Hence

$$0 \leq s_i \leq a - 1, \quad x_i \geq s_i, \quad S \geq r + 2t. \quad (1)$$

The use of actual selections is essential: x_i is not replaced by s_i .

Source demand and supplement forcing

If $ui \rightarrow w$ is selected, every F -neighbour j of i forces an H -edge uj . At most ρ_u such edges are residual. Each other uj is selected, with a distinct supplement w_j different from w . Since ui dominates w_j and u misses w_j , iw_j is an H -edge. It must be residual: i and w_j both miss j in A , whereas a selected cross-edge has its unique exception in B . Distinct supplements give an injection of these remaining F -neighbours into residual edges at i . Thus

$$d_i \leq \rho_u + R_i, \quad \text{and so} \quad s_i \leq \rho_u. \quad (2)$$

Also, the supplement w of $ui \rightarrow w$ neighbours every other selected label j at source u . The quasi-edge uj has a different exception and must dominate w , while u misses w .

2. Exact threshold capacity

For $h \geq 1$ set

$$I_h = \{i : s_i \geq h\}, \quad W_h = \sum_{i \in I_h} s_i, \\ Z_h = \{u : \rho_u \geq h\}, \quad z_h = |Z_h|.$$

Every actual heavy selection has its source in Z_h by (2). A source outside Z_h selects no heavy label; hence all its heavy-label neighbours are residual, and it has at most $h-1$ of them.

If a source has $\ell_u > h$ actual heavy selections, the supplement of each such selection neighbours its other $\ell_u - 1 \geq h$ distinct heavy labels. Such a supplement must lie in Z_h . Let J be these high-load sources and $j = |J|$. Their heavy arcs use distinct unordered pairs inside Z_h incident with J ; there are

$$j(z_h - j) + \binom{j}{2} = j * z_h - j(j+1)/2$$

such pairs. The other $z_h - j$ sources have at most h heavy selections each. Therefore

$$W_h \leq \sum_{u \in Z_h} \ell_u \\ \leq (z_h - j)h + j * z_h - j(j+1)/2.$$

When $W_h > 0$, a heavy label needs at least h distinct selected sources, so $z_h \geq h$. With $q = z_h - h$, the gap from the last bound to $h * z_h + \binom{q}{2}$ is exactly $(q-j)(q-j-1)/2$, nonnegative for every integer $q-j$. Thus

$$2W_h \leq z_h^2 - z_h + h(h+1). \quad (3)$$

If $W_h = 0$ the pointwise argument below is empty. If $H_0 = \max_i s_i > 0$, a label of demand H_0 needs H_0 distinct sources of residual degree at least H_0 . Consequently

$$z_h \geq H_0 \geq h \quad (1 \leq h \leq H_0), \quad \sum_{h=1}^{H_0} z_h \leq r. \quad (4)$$

No graph-wide residual-activity assumption or weak-core reduction is used.

3. Retaining the individual demand profile

For $h \leq H_0$ let $\ell = |I_h|$ and $p = \ell/a$. Every radicand below is positive, since $h \leq s_i \leq a-1$. Cauchy-Schwarz gives

$$\left[(1/a) \sum_{i \in I_h} \sqrt{2a*s_i - h^2} \right]^2 \\ \leq (\ell/a^2) (2a*W_h - \ell*h^2) \\ = 2p*W_h - p^2*h^2.$$

By (3) and $z_h \geq h$, $2W_h \leq z_h^2 + h^2$. Since $0 \leq p \leq 1$ and $z_h \geq h$,

$$2p*W_h - p^2*h^2 \\ \leq p*z_h^2 + p(1-p)h^2 \\ \leq z_h^2.$$

The last gap is $(1-p)(z_h^2 - p*h^2) \geq 0$. Taking nonnegative square roots proves

$$z_h \geq (1/a) \sum_{i: s_i \geq h} \sqrt{2a*s_i - h^2}. \quad (5)$$

Summing (5) and interchanging finite sums, using the shared budget (4), yields

$$r \geq (1/a) \sum_i D_a(s_i), \\ D_a(s) = \sum_{h=1}^s \sqrt{2a*s - h^2}. \quad (6)$$

The source lower bound $z_h \geq h$ is indispensable. Inequality (3) by itself would not justify (5).

4. A rigorously directed sum-to-integral estimate

Define

$$\Phi(x) = \int_0^x \sqrt{2x-y^2} \, dy, \quad 0 \leq x \leq 1.$$

We claim, for integer $0 \leq s \leq a-1$,

$$D_a(s) \geq a^2 \Phi(s/a) - a/4. \quad (7)$$

The case $s=0$ is immediate. Otherwise $a \geq 2$, $s \geq 1$. Let $f(x) = \sqrt{2a*s - x^2}$ on $[0, s+1/2]$. This interval is inside the radicand's nonnegative domain, since

$$\begin{aligned} 2a*s - (s+1/2)^2 - s^2 \\ = 2s(a-s) - s - 1/4 \geq s - 1/4 > 0. \end{aligned}$$

The function is decreasing and concave. For every integer $h=1, \dots, s$, midpoint concavity gives

$$f(h) \geq \int_{h-1/2}^{h+1/2} f(x) \, dx.$$

For completeness, integrate $f(h-u)+f(h+u) \leq 2f(h)$ for $0 \leq u \leq 1/2$. Thus $D_a(s) \geq \int_{1/2}^{s+1/2} f$. The difference between the desired integral over $[0, s]$ and this shifted integral is at most

$$\begin{aligned} [f(0) - f(s+1/2)]/2 \\ \leq [\sqrt{2a*s} - s]/2 \\ \leq a/4. \end{aligned}$$

The middle inequality uses $f(s+1/2) \geq s$, already proved. The final inequality follows from

$$(s+a/2)^2 - 2a*s = (s-a/2)^2 \geq 0.$$

Finally, the substitution $x=ay$ gives $\int_0^s f(x) \, dx = a^2 \Phi(s/a)$, proving (7). In particular, we do NOT incorrectly assert that a decreasing right-endpoint sum is above its unshifted integral; the shift and endpoint correction are essential.

Summing (7) in (6) gives

$$r \geq a \sum_i \Phi(s_i/a) - a/4. \quad (8)$$

5. An elementary scalar certificate; Jensen is not needed

We prove the uniform strict bound

$$x - \Phi(x) < 1/12, \quad 0 \leq x \leq 1. \quad (9)$$

For $0 \leq u \leq 1$,

$$\sqrt{1-u} \geq 1 - u/2 - u^2/2.$$

Indeed $B = 1 - u/2 - u^2/2 = (1-u)(1+u/2) \geq 0$, and

$$(1-u) - B^2 = u^2(1-u)(u+3)/4 \geq 0.$$

For $x > 0$, apply this with $u = y^2/(2x)$ and integrate from 0 to x :

$$\Phi(x) \geq \sqrt{2x} * (x - x^2/12 - x^3/40).$$

Put $q = \sqrt{x/2}$, so $0 \leq q \leq 1/\sqrt{2}$. It suffices to bound

$$P(q) = 2q^2 - 4q^3 + (2/3)q^5 + (2/5)q^7.$$

If $0 \leq q \leq 1/2$, then

$$P(q) \leq 2q^2 - (457/120)q^3.$$

For $c=457/120$, the exact factorisation

$$32 / (27c^2) - (2q^2 - cq^3) = c * (q - 4 / (3c))^2 * (q + 2 / (3c)) \geq 0$$

gives

$$P(q) \leq 51200 / 626547 < 1/12, \\ 1/12 - 51200 / 626547 = 4049 / 2506188 > 0.$$

If $1/2 \leq q \leq 1/\sqrt{2}$, use $q^2 \leq 1/2$ and $q^4 \leq 1/4$ to obtain

$$P(q) \leq 2q^2 - (107/30)q^3.$$

The last cubic is decreasing for $q \geq 1/2$, since its derivative is $q*(4-107q/10) < 0$. Its value at $1/2$ is $13/240 < 1/12$. This proves (9), including $x=0$ directly. Both rational margins are positive; no numerical optimiser or approximate integral is used.

Using (8) and summing (9) gives

$$r > S - a^2/12 - a/4.$$

Combining with $2t \leq S - r$ from (1) proves (P).

Check of the originally proposed Jensen step

For $0 < x \leq 1$,

$$\begin{aligned} \Phi(x) &= (x/2) * \sqrt{2x - x^2} + x * \arcsin(\sqrt{x/2}), \\ \Phi'(x) &= \sqrt{x(2-x)} + \arcsin(\sqrt{x/2}), \\ \Phi''(x) &= (3/2-x) / \sqrt{x(2-x)} > 0. \end{aligned}$$

Continuity at zero makes Φ convex on $[0,1]$, so finite Jensen with weights $1/a$ is valid. It would give $r \geq a^2 * \Phi(S/a^2) - a/4$. However, (9) can be summed directly, making Jensen and these derivative formulae optional rather than dependencies of (T). This removes a potential proof obligation without changing the requested coefficient.

6. Exact degree assembly for $a \geq 31$

Suppose $n \geq 6$, $b \geq 293n/500$ and $m \geq \text{floor}(n^2/4)$. Since $n = a + b + 1$,

$$\begin{aligned} b - n/2 &\geq (43/207)(a+1), \\ t &\geq (b - n/2)^2 - 1/4. \end{aligned}$$

Therefore the difference between this lower bound on t and (P)'s upper bound is at least

$$D(a) = (43/207)^2 * (a+1)^2 - 1/4 - a^2/24 - a/8.$$

Its quadratic coefficient is $509/342792 > 0$. Direct rational arithmetic gives

$$\begin{aligned} D(31) &= 1757/85698 > 0, \\ D(32) - D(31) &= 9401/171396 > 0. \end{aligned}$$

All subsequent forward differences increase by twice the positive quadratic coefficient. Thus $D(a) > 0$ for every integer $a \geq 31$, contradicting (P).

7. Small a , without the older cubic inequality

For $a=0$, G has a universal vertex and edge-criticality forces a star: any edge between other vertices could be deleted without losing diameter two. Thus $m = n - 1 < \text{floor}(n^2/4)$ for $n \geq 6$. For $a=1$, F is edgeless and $t = r \leq 0$; the degree premise with $n \geq 6$ gives a strictly positive required t .

For $2 \leq a \leq 30$ define

$$b_0 = \text{ceil}(293(a+1)/207), \\ T_0 = \text{floor}((b_0 - a - 1)^2/4).$$

The exact identity $\text{floor}(n^{2/4} - b(n-b) = \text{floor}((b-a-1)^2/4)$ incorporates parity. Since $b \geq b_0 > a+1$, the required t is nondecreasing in b . The table in Appendix A shows $24 \cdot T_0 \geq a^2 + 3a$ for every such a except 6 and 11. Those ordinary cases contradict the strict inequality (P).

For the two exceptions we give a small finite integer certificate derived only from (3)-(4). Write $H_0 = \max_{s_i > 0}$ and fix S . For each $1 \leq h \leq H_0$, let k_h be the number of demands at least h . Since

$$S \leq k_h \cdot H_0 + (a - k_h)(h-1),$$

we have

$$K_h = \max(1, \text{ceil}((S - a(h-1))/(H_0 - h + 1))) \leq k_h.$$

Consequently

$$W_h \geq w_h = \max(H_0, h \cdot K_h, S - (a - K_h)(h-1)).$$

Let L_h be the least integer $z \geq H_0$ satisfying $z^2 - z + h(h+1) \geq 2w_h$. The function $z(z-1)$ is increasing on integers $z \geq 1$, so (3)-(4) imply $z_h \geq L_h$ and $r \geq \sum_h L_h$. It follows that

$$S - r \leq S - \sum_h L_h. \quad (10)$$

For fixed a , there are only $1 \leq H_0 \leq a-1$ and $H_0 \leq S \leq a \cdot H_0$ to check. Direct integer evaluation of (10) gives the following row maxima:

	$a=6$: maximum upper bound on	$a=11$: maximum upper bound on
H_0	$S-r$	$S-r$
1	2	6
2	1	8
3	0	7
4	-2	7
5	-7	4
6	not applicable	1
7	not applicable	-4
8	not applicable	-10
9	not applicable	-18
10	not applicable	-27

This is 80 (H_0, S) cases for $a=6$ and 560 for $a=11$, with $290+3,905=4,195$ threshold evaluations. The function `small_certificate` in the supplied standard-library checker reproduces the table using direct integer iteration; it does not call the square-root inversion used by the larger profile audit. The case $S=0$ gives $t \leq 0$ separately.

Thus $a=6$ gives $t \leq 1$, whereas $b_0=10$ requires $t \geq 2$. And $a=11$ gives $t \leq 4$, whereas $b_0=17$ requires $t \geq 6$. These close the two exceptions and finish (T).

In particular, $n=29$ and $\Delta=17$ is excluded from attaining 210 edges by this candidate argument. This does NOT resolve $n=29$ at all maximum degrees or establish a whole-order $n=29$ theorem.

Appendix A. Finite degree comparison

The final column is $24 \cdot T_0 - a^2 - 3a$. Nonnegative values suffice because (P) is strict. The two negative rows use (10) instead.

a	b0	T0	$24 \cdot T0 - a^2 - 3a$
2	5	1	14
3	6	1	6
4	8	2	20
5	9	2	8
6	10	2	-6
7	12	4	26
8	13	4	8
9	15	6	36
10	16	6	14
11	17	6	-10
12	19	9	36
13	20	9	8
14	22	12	50
15	23	12	18
16	25	16	80
17	26	16	44
18	27	16	6
19	29	20	62
20	30	20	20
21	32	25	96
22	33	25	50
23	34	25	2
24	36	30	72
25	37	30	20
26	39	36	110
27	40	36	54
28	42	42	140
29	43	42	80
30	44	42	18

Audit, dependencies and provenance

The starting repository checkpoint was `8e676ae8fac0a7894c649e797a58124ad5a1062d`. The graph-to-threshold antecedent is `../2026-09-08-layer-sum-v1/PROOF.md`, blob `019bf9377d548aaf2990289752c307790149eadb`, with its separate construction, residual-injection and threshold audits. Those arguments are rederived here, not assumed true merely because earlier tests passed. The earlier 13/22 proof, numerical constants and finite-order papers are not changed.

The fresh exact audit examines 478,192 sorted demand profiles through $a=11$ and 4,215,632 associated threshold levels; 833,250 rational midpoint identities; the 640-case small certificate above; and 5,173,536 degree pairs satisfying the proposed ratio with $6 \leq n \leq 5,000$. The full output, parameters and execution environment are saved with the source. These are abstract systems, not actual critical graphs. Some abstract profiles allow positive $S-r$; none is claimed graph-realizable. The finite degree sweep is not a proof of all graphs through 5,000; the universal degree argument is Sections 6-7.

The checker verifies exact identities and integer consequences, not the full calculus or all graph-theoretic steps in a proof assistant. The new result is NOT formally verified. Earlier local Lean lemmas cover only explicitly scoped quasi-edge logic. Independent specialist review, external researcher reproduction and literature priority remain OPEN. No solver UNSAT claim or external endorsement is made.

An official primary documentation cross-check of the optional Jensen statement is Mathlib's `Analy-`

`sis.Convex.Jensen`, in particular `ConvexOn.map_sum_le`. The required equal weights are nonnegative and sum to one, and all s_i/a lie in the proved convexity interval. That documentation is not a formal check of this manuscript. No new best-known literature claim is made.